PREMIER UNIVERSITY

Department of Computer Science & Engineering

Internship Report

A Comparative Study of Neural Network and Deep Learning Models for Sentiment Analysis Using Natural Language Processing (NLP)

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Declaration

I, Fahim Shahriar Mohammad Sakib, a student of the Department of Computer Science and Engineering at Premier University, declare that this internship report titled “A Comparative Study of Neural Network and Deep Learning Models for Sentiment Analysis Using Natural Language Processing” has been prepared to document my research and learning activities during my internship with Authentic Four Technology from 1 September 2026 to 20 September 2026. The experimental work, results, and figures presented in this report were obtained from the research implementation carried out in Google Colab, and, to the best of my knowledge, they accurately reflect the work actually completed. Any external material used for background and technical explanation is acknowledged in the references.

Signature: _____ Date: _____

Acknowledgement

I would like to express my sincere gratitude to Authentic Four Technology for providing me with the opportunity to undertake a research-oriented internship. The internship introduced me to a structured research workflow and helped me connect concepts from computer science with a practical problem: comparing neural network and deep learning models for sentiment analysis using natural language processing.

I am grateful to my academic supervisor, Ms. Farhana Shirin Chowdhury, Associate Professor, Department of Computer Science and Engineering, Premier University, for academic guidance and support. I also thank my industry mentor, Imtiaz Uddin Riad, Chief Technology Officer, Authentic Four Technology, and everyone at the organization who contributed to the internship activities and learning environment.

I would also like to acknowledge the contributor of the publicly available IMDb Dataset of 50K Movie Reviews on Kaggle, and the developers of the Python, TensorFlow, Keras, Scikit-learn, and Google Colab tools used in the research workflow. Finally, I thank my department and university for making this internship experience part of my academic development.

Abstract

Sentiment analysis is an important application of Natural Language Processing (NLP) that aims to identify and classify opinions expressed in textual data. With the increasing availability of online reviews, automatically identifying positive and negative opinions has become valuable for understanding customer feedback and public opinion.

This study presents a comparative analysis of four neural network and deep learning architectures for binary sentiment classification: Artificial Neural Network (ANN), Simple Recurrent Neural Network (SimpleRNN), Long Short-Term Memory (LSTM), and Bidirectional Long Short-Term Memory (BiLSTM). The IMDb 50K Movie Reviews dataset, containing 50,000 reviews with equal numbers of positive and negative samples, was used for the experiment.

The dataset underwent several preprocessing steps, including lowercase conversion, HTML tag removal, URL removal, special character removal, tokenization, and sequence padding. The data was divided into training and testing sets using an 80:20 ratio. Each model was trained and evaluated using the same dataset split. Performance was measured using accuracy, precision, recall, and F1-score.

The experimental results showed that ANN achieved the highest accuracy of 88.98% and the highest F1-score of 88.70%. BiLSTM achieved the highest precision of 92.36%, while SimpleRNN and LSTM demonstrated comparatively lower performance under the selected experimental configurations.

The findings indicate that model complexity does not necessarily guarantee better classification performance. This study provides a comparative understanding of different neural network architectures for sentiment analysis and identifies areas for further improvement.

Keywords: Sentiment Analysis, Natural Language Processing, Artificial Neural Network, SimpleRNN, LSTM, BiLSTM, Deep Learning, IMDb Movie Reviews.

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1 Introduction

1.1 Background

In recent years, the rapid growth of digital platforms has resulted in a significant increase in the amount of textual data generated by users. People regularly share their opinions, experiences, and feedback through online reviews, social media platforms, blogs, and discussion forums.

Understanding this textual information is important for businesses, researchers, and organizations. However, manually analyzing a large volume of text is time-consuming and difficult. Natural Language Processing (NLP) provides techniques that allow computers to process, understand, and analyze human language.

One important application of NLP is sentiment analysis. Sentiment analysis is the process of identifying the emotional tone or opinion expressed in a piece of text. In binary sentiment classification, textual data is classified into two categories: positive and negative. For example:

- "This movie was amazing and entertaining." — Positive
- "The movie was boring and disappointing." — Negative

Machine learning and deep learning techniques have become widely used for sentiment classification. Neural network architectures can learn patterns from textual data and use those patterns to predict sentiment categories.

1.2 Problem Statement

Although several neural network architectures are available for text classification, their performance may differ depending on the dataset, preprocessing techniques, model architecture, and training configuration.

Traditional feed-forward neural networks, recurrent neural networks, and advanced recurrent architectures such as LSTM and BiLSTM process textual information differently. Therefore, it is important to investigate how these architectures perform when applied to the same sentiment classification task under a consistent experimental setup.

This research focuses on comparing four neural network architectures to identify their strengths and limitations in classifying positive and negative movie reviews.

1.3 Objectives

The main objectives of this research are:

1. To understand the application of Natural Language Processing in sentiment analysis.
2. To explore and preprocess the IMDb 50K Movie Reviews dataset.
3. To implement four neural network architectures: ANN, SimpleRNN, LSTM, and BiLSTM.
4. To evaluate the models using accuracy, precision, recall, and F1-score.

5. To compare the performance of the selected models using quantitative results and visualization.
6. To identify the best-performing model within the experimental setup.

1.4 Research Questions

This study aims to answer the following research questions:

1. How effectively can neural network models classify positive and negative movie reviews?
2. How does the performance of ANN compare with SimpleRNN, LSTM, and BiLSTM?
3. Which model achieves the highest accuracy and F1-score?
4. Does a more complex recurrent architecture necessarily provide better sentiment classification performance?
5. What limitations affect the performance of the evaluated models?

1.5 Scope of the Research

This research focuses on binary sentiment classification using English-language movie reviews. The study uses the IMDb 50K dataset and evaluates four neural network architectures.

The research is limited to identifying positive and negative sentiment. Neutral sentiment classification, multilingual sentiment analysis, and transformer-based models are outside the scope of this experiment.

1.6 Report Organization

The report follows the suggested internship report structure: introduction, company profile, organizational structure, technologies used, internship activities, technical learning, problems observed, proposed solutions, reflection on professional skills, conclusion, references, and appendices. The detailed experimental results and their interpretation are presented in Appendices B and C.

2 Company Profile

2.1 Organization Overview

Authentic Four Technology is the organization where the internship was undertaken. According to its official website, it describes itself as a leading software company in Bangladesh that offers customized or ready-made mobile app, web, and software solutions. The company's official website is https://www.authenticfour.com

2.2 Services and Areas of Work

The company's website lists the following services:

- **App Development:** mobile app development solutions for major platforms, including iPhone and Android.
- **Software Development:** software solutions, including ERP software.
- **Web Development:** web design and development.
- **WordPress Development:** website development on the WordPress platform.
- **E-commerce Website:** design and development of e-commerce websites.
- **Digital Marketing:** web advertising and digital marketing solutions.

The website also presents ready-made software solutions, including Sales ERP, Telemedicine Mobile App, Hospital Management, Inventory Management, Pharmacy Management, Human Resource Management, and Restaurant Management software.

2.3 Relevance to the Internship

The internship assignment was research-focused. It concerned a comparative study of neural network and deep learning models for sentiment analysis using NLP, carried out in the Google Colab environment on a public dataset. The internship connected data science concepts with practical activities such as dataset inspection, text preprocessing, model development, evaluation, and documentation of experiments. As a software and technology organization, Authentic Four Technology provided a suitable environment for understanding how data-driven and AI-related ideas can be translated into practical computational workflows.

2.4 Official Website

The official website of Authentic Four Technology is https://www.authenticfour.com. It may be consulted for the latest company description, services, and branding.

3 Organizational Structure

3.1 Internship Supervision

The academic supervisor for this internship is Ms. Farhana Shirin Chowdhury, Associate Professor, Department of Computer Science and Engineering, Premier University. The organization also appointed an industry mentor, Imtiaz Uddin Riad, Chief Technology Officer, Authentic Four Technology.

3.2 Communication and Responsibilities

The intern's responsibilities included understanding the assigned research topic, obtaining and exploring the IMDb 50K Movie Reviews dataset, preprocessing the text data, implementing and training four neural network architectures, evaluating the models, documenting the experiments in a Google Colab notebook, and preparing this report.

4 Technologies Used

4.1 Python

Python is the programming language used for the research implementation. Its readable syntax and extensive ecosystem of data science and machine learning libraries make it suitable for exploratory research. In this study, Python was used to load and inspect data, preprocess text, build and train neural networks, and calculate evaluation metrics.

4.2 Google Colab

Google Colab is a cloud-hosted notebook environment that allows Python code to be executed in interactive cells. Code, explanatory notes, outputs, and visualizations can be kept together, which is useful for research. All experiments in this study were developed and executed in Google Colab. The notebook, named `IMDb_50K_Internship_Research.ipynb`, contains the complete workflow, including data loading, preprocessing, exploratory data analysis, model development, model training, evaluation, and visualization.

4.3 Kaggle Dataset Platform

Kaggle is a platform for hosting and sharing datasets and machine-learning projects. The IMDb 50K Movie Reviews dataset used in this study was obtained from Kaggle. It contains 50,000 movie reviews equally divided between positive and negative sentiment. The dataset link is provided in Appendix A.

4.4 Python Libraries and Frameworks

The following libraries and frameworks were used in the experimental workflow:

Table 1: Libraries and frameworks used in the experiments.

Tool / Library	Purpose
Pandas	Dataset loading and manipulation
NumPy	Numerical operations
Matplotlib	Data visualization
Seaborn	Statistical visualization
Scikit-learn	Dataset splitting and evaluation
TensorFlow / Keras	Neural network development and training
WordCloud	Text visualization

4.5 NLP and Neural Network Concepts

This section summarizes the concepts and prior work on which the experimental study is based.

4.5.1 Sentiment Analysis Using Natural Language Processing

Sentiment analysis is an important research area in Natural Language Processing. It focuses on identifying opinions, emotions, and subjective information from textual data.

Maas et al. (2011) [1] introduced a large movie review dataset and investigated learning word representations that capture semantic and sentiment information. Their research established an important benchmark for document-level sentiment classification. The IMDb movie review dataset has subsequently been used in numerous sentiment analysis experiments to evaluate different text classification approaches.

4.5.2 Artificial Neural Networks

Artificial Neural Networks are computational models inspired by the structure and functioning of biological neural systems. They consist of interconnected processing units, commonly organized into input, hidden, and output layers. ANN models learn patterns by adjusting their weights during training, and they are widely used in classification and prediction tasks.

In text classification, word embeddings can represent words as numerical vectors. These representations can then be processed by neural network layers to predict the sentiment of a review. In this study, an embedding-based ANN architecture was implemented using an Embedding layer, Global Average Pooling, a Dense hidden layer, Dropout, and a sigmoid output layer.

4.5.3 Recurrent Neural Networks

Recurrent Neural Networks are designed to process sequential data. Unlike conventional feed-forward networks, RNNs maintain information from previous time steps through recurrent connections. Elman (1990) [2] investigated recurrent connections as a mechanism for representing temporal information and learning structures in sequential data.

In sentiment analysis, RNNs can process words in sequence and learn relationships between words within a review. However, standard recurrent networks can experience difficulties learning long-range dependencies because information and gradients may become difficult to preserve over long sequences. This study uses a SimpleRNN architecture to investigate the performance of a basic recurrent model on movie review classification.

4.5.4 Long Short-Term Memory Networks

Long Short-Term Memory networks were introduced by Hochreiter and Schmidhuber (1997) [3] to address difficulties associated with learning long-term dependencies in recurrent neural networks. LSTM uses a memory cell and gating mechanisms to regulate the flow of information. These mechanisms help the network retain or discard information during sequential processing.

Because movie reviews may contain sentiment-related information across long sequences of words, LSTM is a relevant architecture to investigate for sentiment classification. In this experiment, an LSTM layer was used to process tokenized movie reviews and predict their sentiment.

4.5.5 Bidirectional LSTM Networks

Bidirectional Recurrent Neural Networks were introduced by Schuster and Paliwal (1997) [4]. Unlike conventional unidirectional recurrent networks, bidirectional networks process sequential information in both forward and backward directions. A Bidirectional LSTM combines the memory capabilities of LSTM with forward and backward sequence processing, allowing the model to consider information from both directions when learning contextual relationships.

For sentiment analysis, considering context from both directions may help identify relationships between words and their surrounding expressions. This research implements a Bidirectional LSTM model to compare its performance with ANN, SimpleRNN, and conventional LSTM.

4.5.6 Research Gap and Motivation

Previous research has established the importance of neural network architectures and word representations for sentiment classification. However, model performance depends on the dataset, preprocessing procedures, architecture, and training configuration.

This study provides an experimental comparison of four neural network architectures using the same IMDb movie review dataset, training-test split, and evaluation metrics. The motivation is to investigate whether more complex recurrent architectures necessarily outperform a simpler embedding-based neural network for binary movie review sentiment classification.

5 Internship Activities

5.1 Internship Overview

The internship was undertaken from 1 September 2026 to 20 September 2026 (20days). The assigned research topic was a comparative study of four neural network architectures (ANN, SimpleRNN, LSTM, and BiLSTM) for binary sentiment classification of movie reviews, using the IMDb 50K Movie Reviews dataset in Google Colab.

5.2 Summary of Research Activities

The research was carried out in the following stages. Each stage is described in more detail in Section 6.

Table 2: Summary of research activities.

Stage	Activity and outcome
1	Dataset collection: the IMDb 50K Movie Reviews dataset was obtained from Kaggle (50,000 reviews; 25,000 positive and 25,000 negative).
2	Exploratory data analysis: dataset dimensions, missing values, duplicate records (418 identified), sentiment distribution, review lengths, and word clouds were examined.
3	Text preprocessing: lowercase conversion, HTML tag removal, URL removal, special character removal, and whitespace normalization; cleaned text stored in a new column.

Stage	Activity and outcome
4	Tokenization and sequence padding: Keras Tokenizer with a 20,000-word vocabulary and a maximum sequence length of 300 tokens.
5	Data preparation: stratified 80:20 split into 40,000 training and 10,000 testing reviews (random state 42).
6	Model development: ANN, SimpleRNN, LSTM, and BiLSTM models implemented using TensorFlow and Keras.
7	Model training and validation: Adam optimizer, binary cross-entropy loss, up to 10 epochs, batch size 128, 10% validation split, and Early Stopping.
8	Model evaluation: accuracy, precision, recall, F1-score, confusion matrices, and classification reports.
9	Comparative analysis and documentation: results compared using tables and visualizations; limitations identified; report prepared.

5.3 Research Materials

The principal research resources used in this internship are:

1. **Dataset:** IMDb 50K Movie Reviews dataset from Kaggle (link in Appendix A).
2. **Google Colab notebook:** `IMDb_50K_Internship_Research.ipynb`, containing the complete experimental workflow.
3. **Saved models:** the four trained models were saved as `IMDb_ANN_Model.keras`, `IMDb_RNN_Model.keras`, `IMDb_LSTM_Model.keras`, and `IMDb_BiLSTM_Model.keras`.

6 Technical Learning

This section describes the research design and the technical methods applied in the experiments. This research follows an experimental approach to compare four neural network architectures for binary sentiment classification. All four models were evaluated using the same initial training and testing split to maintain consistency in the comparison.

6.1 Understanding the Dataset

The IMDb 50K Movie Reviews dataset was obtained from Kaggle. It contains 50,000 movie reviews with two sentiment categories: positive and negative. The dataset was balanced, containing an equal number of positive and negative reviews. The original dataset contained

two columns: `review` and `sentiment`. Additional columns were created during preprocessing and analysis.

Table 3: Dataset information.

Dataset Information	Value
Total Reviews	50,000
Positive Reviews	25,000
Negative Reviews	25,000
Original Columns	2
Duplicate Rows Identified	418

6.2 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the dataset before model development. The following activities were conducted:

- Examining dataset dimensions and column information.
- Checking missing values and duplicate records.
- Analyzing positive and negative sentiment distribution.
- Calculating the length of movie reviews.
- Visualizing review length distributions.
- Comparing review lengths between sentiment categories.
- Generating word clouds for positive and negative reviews.

These analyses provided an initial understanding of the dataset and its textual characteristics. A total of 418 duplicate rows were identified during data inspection. However, these duplicate records were not removed before the initial train-test split and model training. The EDA results are presented in Appendix B.

6.3 Text Preprocessing

Text preprocessing was performed to convert raw movie reviews into a more consistent format suitable for neural network training. The following preprocessing operations were applied:

1. **Lowercase Conversion:** All text was converted into lowercase letters to maintain consistency.
2. **HTML Tag Removal:** HTML tags were removed from the review text.
3. **URL Removal:** Website links and URLs were removed.
4. **Special Character Removal:** Non-alphabetic characters were replaced with spaces.
5. **Whitespace Normalization:** Multiple spaces were replaced with a single space, and unnecessary spaces were removed.

The cleaned reviews were stored in a new column named `cleaned_review`. The original review text was preserved for comparison and analysis.

The sentiment labels were also converted into numerical values because neural network models require numerical target values. The encoded values were stored in the `sentiment_encoded` column.

Table 4: Sentiment label encoding.

Sentiment	Encoded Value
Negative	0
Positive	1

6.4 Tokenization and Sequence Padding

After text cleaning, tokenization was performed using the Keras Tokenizer. The tokenizer converted words into numerical integer sequences based on their frequency in the training dataset. The vocabulary size was limited to 20,000 words, and an out-of-vocabulary token was included to handle words not present in the learned vocabulary. The maximum sequence length was set to 300 tokens. Reviews shorter than the maximum length were padded with zeros, while longer reviews were truncated.

Table 5: Tokenization and sequence parameters.

Parameter	Value
Maximum Vocabulary Size	20,000
Maximum Sequence Length	300
Embedding Dimension	64
Padding	Post
Truncating	Post
OOV Token	`<OOV>`

6.5 Training and Testing Data Split

The dataset was divided into training and testing subsets using an 80:20 ratio. The `train_test_split` function from Scikit-learn was used with stratified sampling to preserve the sentiment distribution across both subsets.

Table 6: Training and testing data split.

Dataset Subset	Number of Reviews	Percentage
Training Set	40,000	80%
Testing Set	10,000	20%
Total	50,000	100%

The random state was set to 42 to make the split reproducible. The tokenizer was fitted only on the training text to reduce vocabulary information leakage from the testing set.

6.6 Model Architectures

Four neural network architectures were developed using TensorFlow and Keras.

6.6.1 Artificial Neural Network (ANN)

The ANN model consisted of the following layers:

- Embedding layer with 64 dimensions.
- Global Average Pooling 1D layer.
- Dense hidden layer with 64 neurons and ReLU activation.
- Dropout layer with a rate of 0.5.
- Output Dense layer with one neuron and sigmoid activation.

Global Average Pooling converted the sequence of word embeddings into a fixed-size representation by averaging across the sequence positions.

6.6.2 Simple Recurrent Neural Network (SimpleRNN)

The SimpleRNN architecture consisted of:

- Embedding layer with 64 dimensions.
- SimpleRNN layer with 64 units and tanh activation.
- Dropout layer with a rate of 0.5.
- Output Dense layer with one neuron and sigmoid activation.

This model was designed to learn sequential patterns from the movie reviews.

6.6.3 Long Short-Term Memory (LSTM)

The LSTM architecture consisted of:

- Embedding layer with 64 dimensions.
- LSTM layer with 64 units.
- Dropout layer with a rate of 0.5.
- Output Dense layer with one neuron and sigmoid activation.

The LSTM layer was used to learn sequential information and retain relevant contextual information during processing.

6.6.4 Bidirectional LSTM (BiLSTM)

The BiLSTM architecture consisted of:

- Embedding layer with 64 dimensions.
- Bidirectional LSTM layer with 64 units in each direction.
- Dropout layer with a rate of 0.5.
- Output Dense layer with one neuron and sigmoid activation.

The Bidirectional layer processed the input sequence in both forward and backward directions.

6.7 Model Training Configuration

All four models were compiled using the Adam optimizer and binary cross-entropy loss function.

Table 7: Model training configuration.

Training Parameter	Value
Optimizer	Adam
Loss Function	Binary Cross-Entropy
Training Metric	Accuracy
Maximum Epochs	10
Batch Size	128
Validation Split	10% of Training Data
Dropout Rate	0.5
Early Stopping Patience	2
Restore Best Weights	True

Early Stopping was applied by monitoring validation loss. Training was stopped when the validation loss failed to improve for two consecutive epochs, and the best model weights were restored.

6.8 Evaluation Metrics

The performance of each model was evaluated using four standard classification metrics: accuracy, precision, recall, and F1-score. Their definitions are given in Table 8.

Table 8: Definitions of the evaluation metrics.

Metric	Formula	Meaning
Accuracy	(TP + TN) / (TP + TN + FP + FN)	Proportion of correctly classified reviews among all tested reviews.
Precision	TP / (TP + FP)	Proportion of predicted positive reviews that were actually positive.
Recall	TP / (TP + FN)	Proportion of actual positive reviews correctly identified by the model.
F1-Score	2 × (Precision × Recall) / (Precision + Recall)	Harmonic mean of precision and recall.

Here, TP = True Positive, TN = True Negative, FP = False Positive, and FN = False Negative. Confusion matrices and classification reports were also generated to examine the classification performance of each model.

6.9 Research Workflow

Figure 1 presents the overall workflow followed in this research, from dataset collection to comparative analysis.

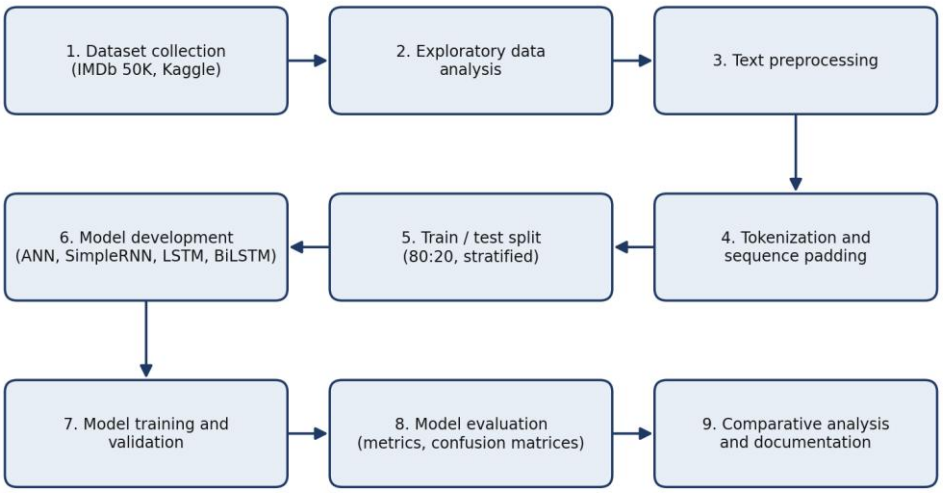


Figure 1: Research workflow for the comparative sentiment analysis experiments.

7 Problems Observed

The following problems and limitations were observed during the research and should be considered when interpreting the findings.

7.1 Duplicate Records

The original dataset contained 418 duplicate rows. These records were not removed before splitting the dataset. Consequently, identical review records may have appeared in both training and testing subsets. This creates a possibility of data leakage and should be considered when interpreting the experimental results.

7.2 Poor Performance of SimpleRNN and LSTM

SimpleRNN achieved an accuracy of only 50.62%, and LSTM achieved an accuracy of 55.48% with a recall of 12.94%. SimpleRNN classified many negative reviews as positive, while LSTM failed to identify many positive reviews. This suggests that these architectures and their training configurations may require further optimization.

7.3 Model Configuration

The models were trained using specific hyperparameters, including a maximum vocabulary size of 20,000, a sequence length of 300, an embedding dimension of 64, and a batch size of 128. Different configurations may produce different results.

7.4 Limited Model Architectures

Only four neural network architectures were evaluated. Other approaches, such as GRU, CNN, and Transformer-based models, were not included.

7.5 Sequence Truncation

Reviews longer than 300 tokens were truncated, meaning that some contextual information may have been lost.

7.6 Computational Constraints

The experiments were conducted using a particular computing environment. Training time and performance may vary across hardware configurations.

8 Proposed Solutions

Several improvements can be explored in future research to address the problems observed and to enhance sentiment analysis performance.

8.1 Remove Duplicate Data

Future experiments should remove duplicate reviews before splitting the dataset to reduce the possibility of data leakage and obtain more reliable evaluation results.

8.2 Hyperparameter Optimization

Different learning rates, batch sizes, embedding dimensions, vocabulary sizes, and sequence lengths can be tested to improve model performance.

8.3 Additional Deep Learning Models

Other architectures, such as Gated Recurrent Unit (GRU), Convolutional Neural Networks (CNN), and Transformer-based models, can be investigated.

8.4 Advanced NLP Techniques

More advanced text preprocessing, word embeddings, and contextual language representations can be explored to improve sentiment understanding.

8.5 Larger and More Diverse Datasets

Future studies can use reviews from different domains, such as product reviews, restaurant feedback, and social media comments, to evaluate model generalization.

8.6 Model Efficiency

Future research can compare training time, computational requirements, and model size alongside classification metrics.

These improvements may provide more reliable results and help develop more effective sentiment analysis systems.

9 Reflection on Professional Skills

Through this research project, I gained practical experience in Natural Language Processing (NLP), Neural Networks, and Deep Learning. I worked with the IMDb 50K Movie Reviews dataset and implemented four different models: ANN, SimpleRNN, LSTM, and BiLSTM. This experience helped me understand text preprocessing, model training, performance evaluation, and comparative analysis. I also learned how different neural network architectures perform in sentiment classification and improved my practical skills in Python, TensorFlow, and Google Colab. Overall, this project strengthened my understanding of AI-based text analysis and research methodology.

9.1 Technical Skills

The internship provided practical experience in a complete deep learning workflow for text data: loading and inspecting a dataset, cleaning text, tokenization and sequence padding, building ANN, SimpleRNN, LSTM, and BiLSTM models using TensorFlow and Keras, training with Early Stopping, and evaluating models using accuracy, precision, recall, F1-score, and confusion matrices.

9.2 Research Skills

The experiment was designed so that all four models used the same dataset split and evaluation metrics, allowing a fair comparison. The results showed that a more complex architecture does not necessarily perform better, and that evaluation should consider more than one metric. Identifying the duplicate records and reporting them as a limitation reinforced the importance of examining data quality and reporting findings honestly.

9.3 Documentation and Communication

The research was documented in a Google Colab notebook, the trained models were saved, and the results were organized into tables and figures. Preparing this report also developed the ability to explain technical concepts and experimental findings in a structured way for academic readers.

9.4 Professional Ethics

Professional responsibility in research includes acknowledging limitations and avoiding overstated claims. In this study, the findings were stated as specific to the experimental setup and were not generalized to all sentiment analysis tasks.

9.5 Future Development

The experience encouraged further study of hyperparameter optimization, additional architectures such as GRU, CNN, and Transformer-based models, and evaluation on deduplicated and more diverse datasets.

10 Conclusion

This research investigated the performance of different neural network and deep learning architectures for sentiment analysis using Natural Language Processing. The IMDb 50K Movie Reviews dataset was used to classify movie reviews into positive and negative sentiment categories. Four models were implemented and evaluated: Artificial Neural Network (ANN), Simple Recurrent Neural Network (SimpleRNN), Long Short-Term Memory (LSTM), and Bidirectional LSTM (BiLSTM).

The experimental results showed that ANN achieved the highest accuracy of 88.98% and an F1-score of 88.70%. BiLSTM achieved an accuracy of 88.05% and the highest precision of 92.36%. In comparison, SimpleRNN and LSTM demonstrated lower classification performance under the selected experimental configurations.

The findings indicate that model complexity alone does not guarantee better sentiment classification performance. The effectiveness of a model depends on its architecture, preprocessing techniques, and training configuration.

Overall, ANN demonstrated the best overall performance in this experiment. However, the presence of duplicate records in the original dataset represents a limitation, and further validation using a deduplicated dataset is recommended. This study provides a comparative understanding of neural network architectures and their application to movie review sentiment analysis.

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12 Appendices

12.1 Appendix A: Key Research Links and Materials

- **Dataset:** IMDb Dataset of 50K Movie Reviews (Kaggle) - https://www.kaggle.com/datasets/lakshmi25npathi/imdb-dataset-of-50k-movie-reviews
- **Google Colab notebook:** https://colab.research.google.com/drive/1y0zbCQ-ryf54F03u_5gWqAMouycxNrnG#scrollTo=CaXRdHvaEbTf
- **Saved trained models:** `IMDb_ANN_Model.keras`, `IMDb_RNN_Model.keras`, `IMDb_LSTM_Model.keras`, and `IMDb_BiLSTM_Model.keras`.

12.2 Appendix B: Experimental Results

This appendix presents the experimental results obtained from the research implementation in Google Colab. All four models were evaluated on the same testing set of 10,000 movie reviews.

12.2.1 Dataset Analysis

The IMDb dataset contained 50,000 movie reviews, equally distributed between positive and negative categories (Figure 2). During data exploration, 418 duplicate rows were identified. The dataset contained no missing values in the original review and sentiment columns. The dataset was divided into 40,000 training samples and 10,000 testing samples.

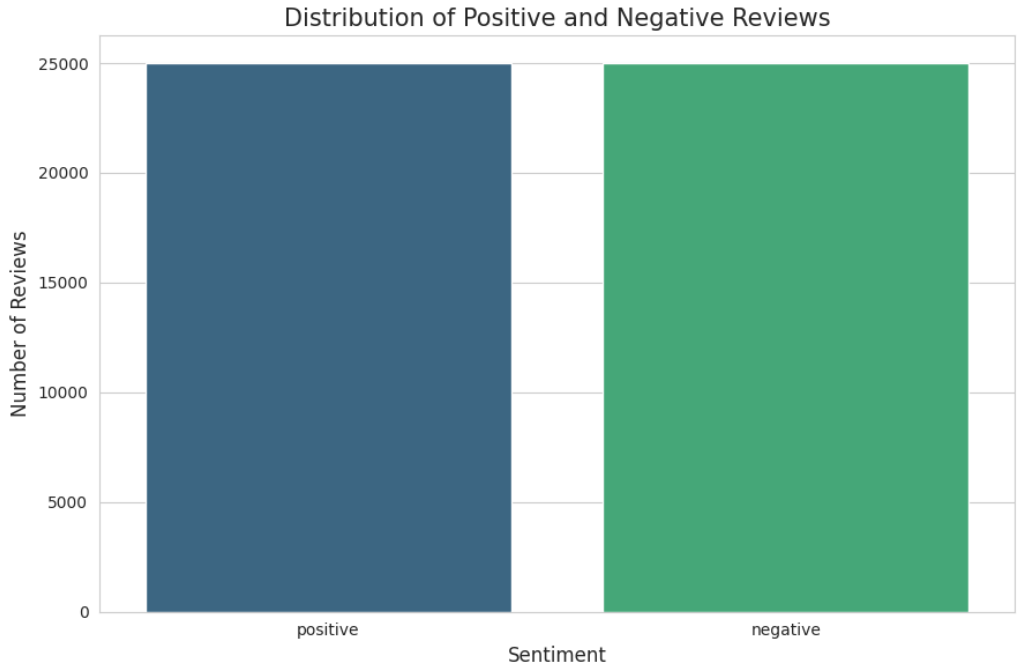


Figure 2: Sentiment distribution.

The length of each movie review was calculated by counting the number of words in the review. A histogram was generated to visualize the distribution of review lengths (Figure 3), and a boxplot was used to compare review lengths between positive and negative categories (Figure 4).

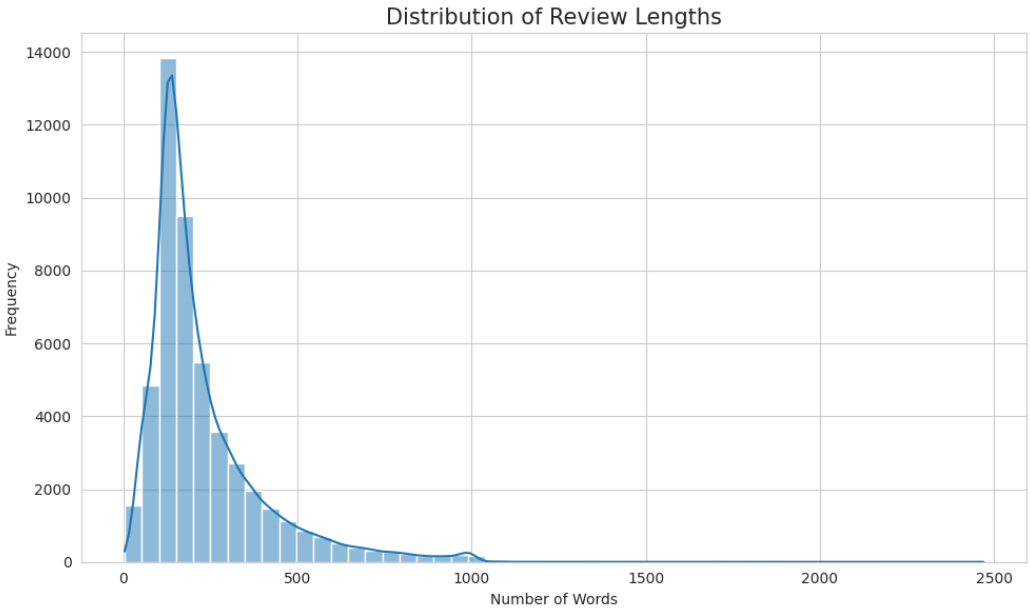


Figure 3: Review length distribution.



Figure 4: Review length by sentiment.

Word clouds were generated separately for positive and negative reviews to visualize frequently occurring words in each sentiment category (Figures 5 and 6). These visualizations provided an overview of common vocabulary used in positive and negative movie reviews.



Figure 5: Positive review word cloud.

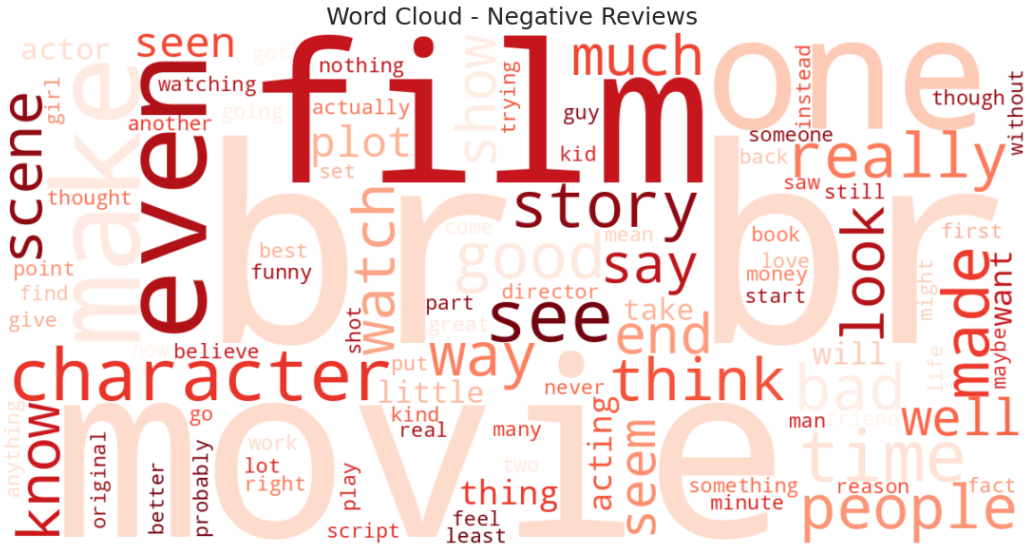


Figure 6: Negative review word cloud.

12.2.2 Model Training Results

The four neural network models were trained using the same training dataset and validation configuration. Early Stopping was applied to monitor validation loss. The number of completed training epochs for each model is shown in Table 9, and the training curves are shown in Figures 7 to 10.

Table 9: Completed training epochs.

Model	Completed Epochs
ANN	6
SimpleRNN	7
LSTM	4
BiLSTM	5

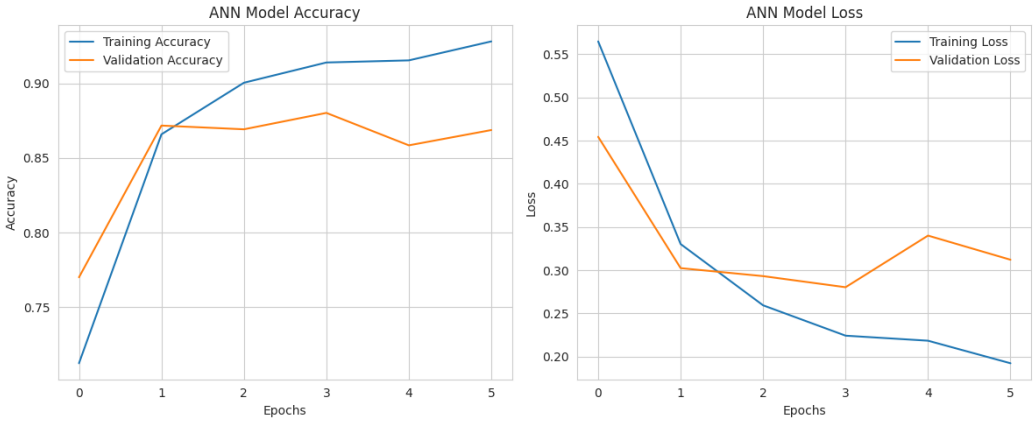


Figure 7: ANN training accuracy and loss.

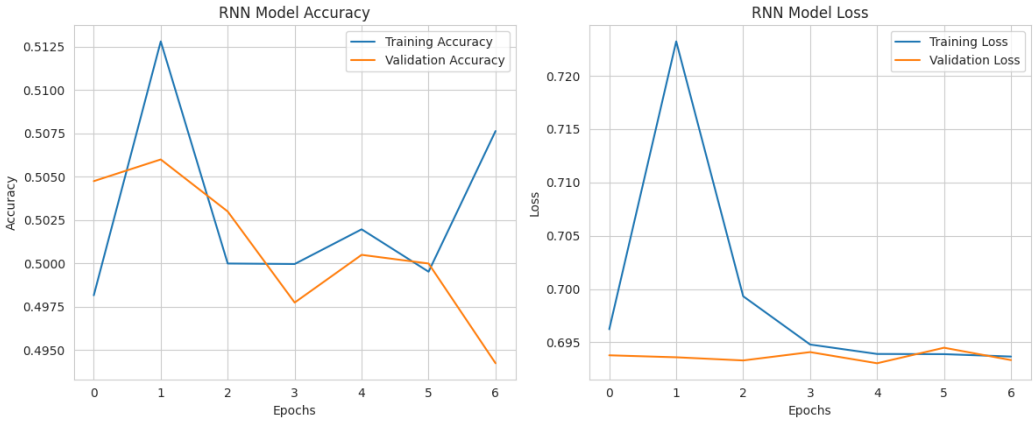


Figure 8: SimpleRNN training accuracy and loss.

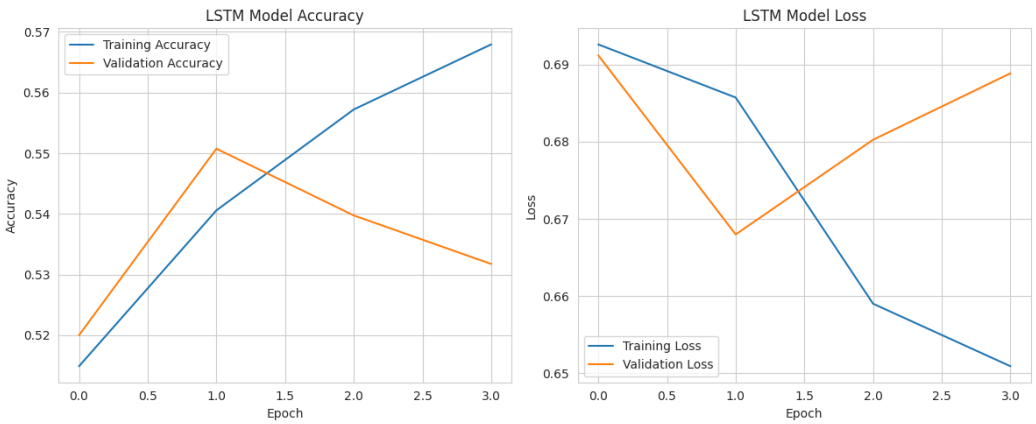


Figure 9: LSTM training accuracy and loss.

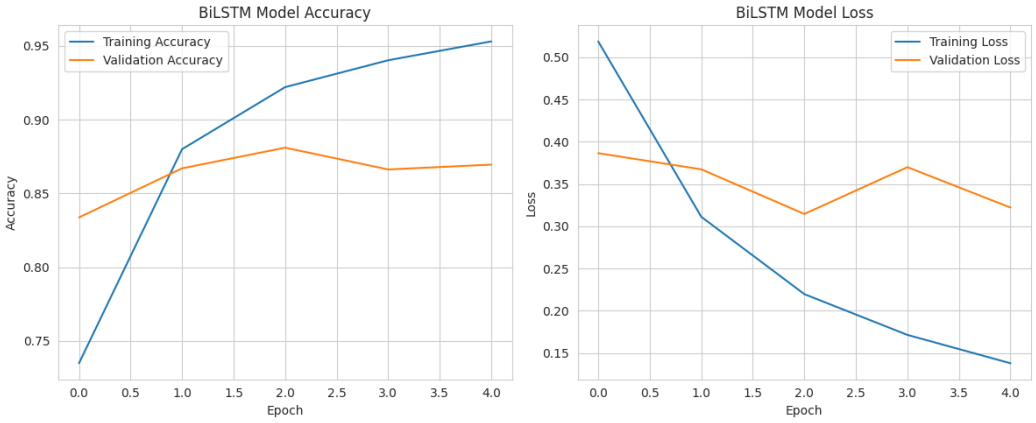


Figure 10: BiLSTM training accuracy and loss.

12.2.3 Comparative Model Performance

The evaluation results on the testing dataset are presented in Table 10. The highest value in each metric column is shown in bold.

Table 10: Comparative performance of the four models on the testing set (10,000 reviews).

Model	Accuracy	Precision	Recall	F1-Score
ANN	**88.98%**	91.03%	**86.48%**	**88.70%**
SimpleRNN	50.62%	50.40%	78.66%	61.43%
LSTM	55.48%	86.73%	12.94%	22.52%
BiLSTM	88.05%	**92.36%**	82.96%	87.41%

The ANN model achieved the highest accuracy and F1-score. BiLSTM achieved the highest precision.

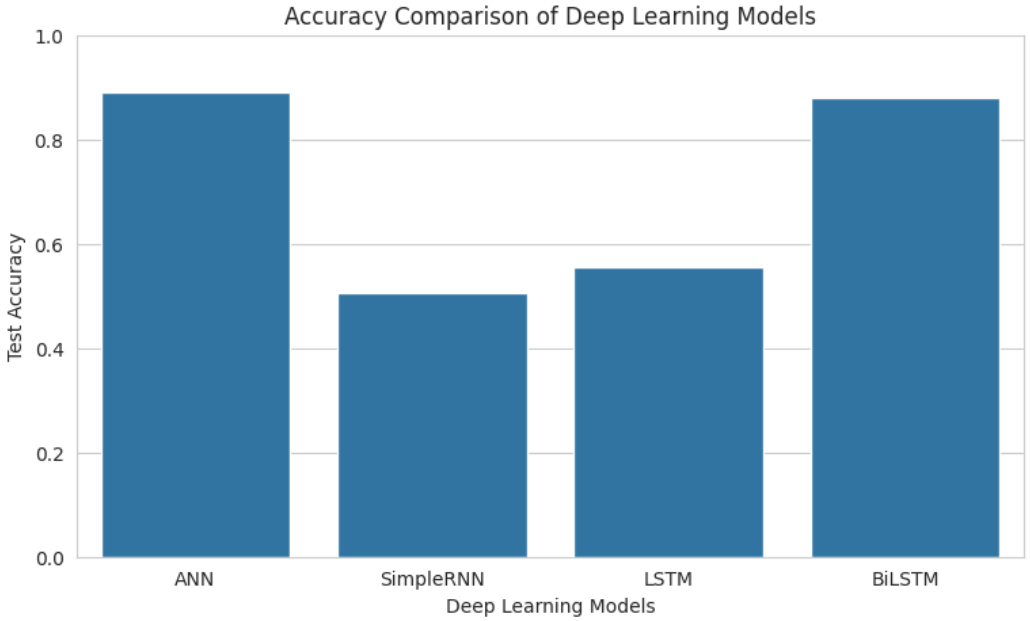


Figure 11: Model accuracy comparison.

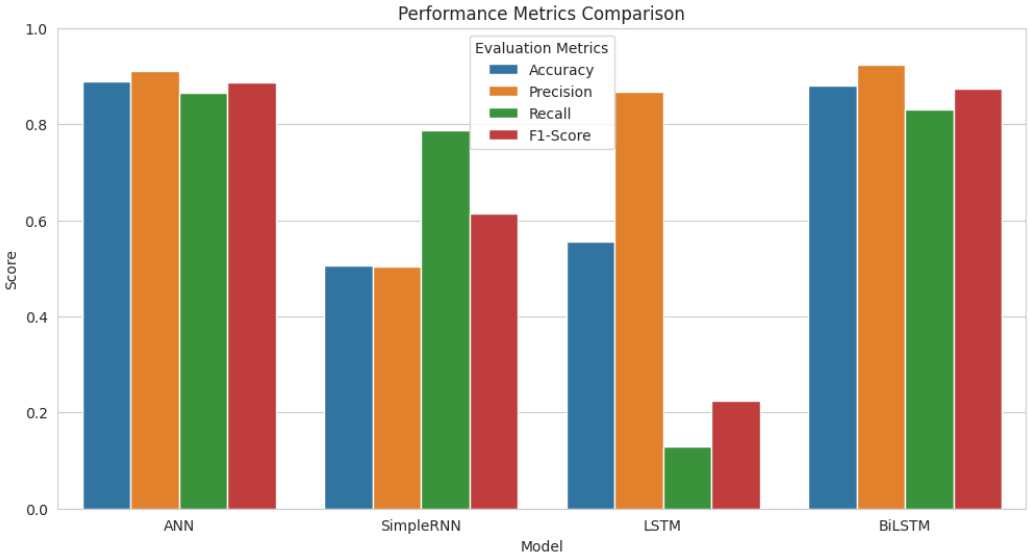


Figure 12: All model metrics comparison.

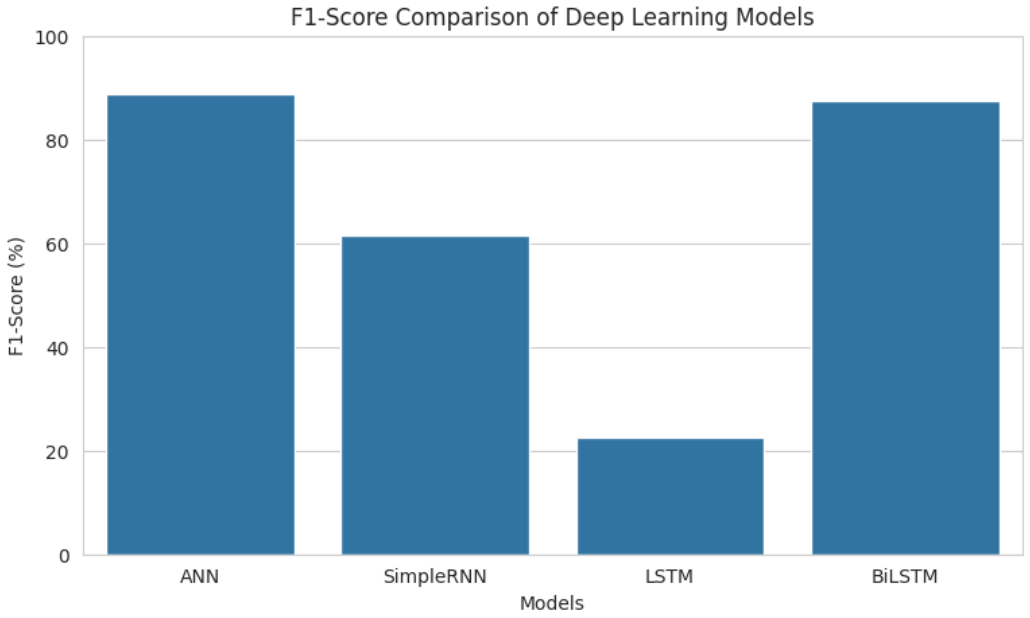


Figure 13: F1-score comparison.

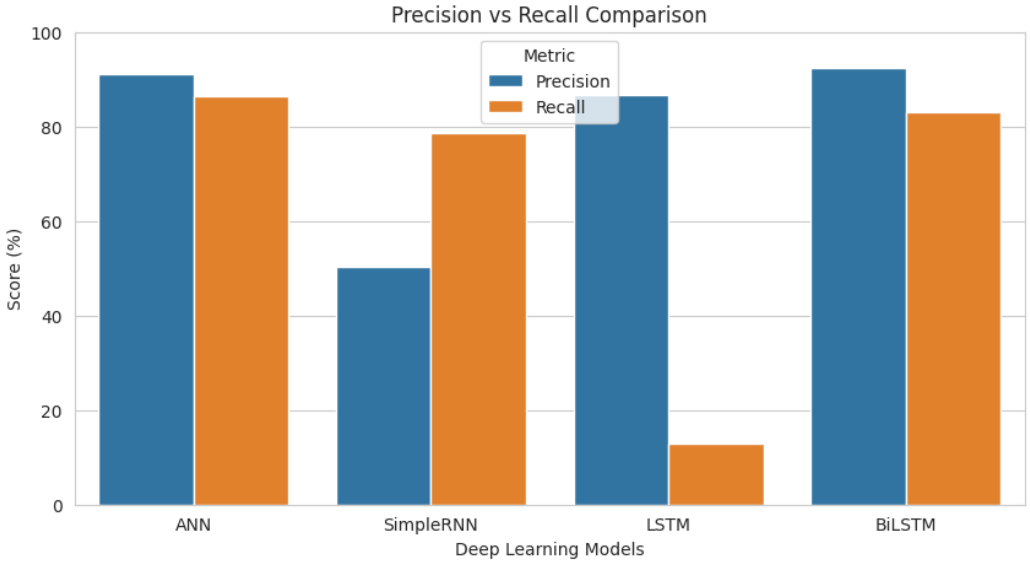


Figure 14: Precision and recall comparison.

12.2.4 Confusion Matrices

The confusion matrices provided additional information about the classification performance of each model. The counts are given in Table 11 and visualized in Figures 15 to 19.

Table 11: Confusion matrix counts for each model.

Model	True Negative	False Positive	False Negative	True Positive
ANN	4,574	426	676	4,324
SimpleRNN	1,129	3,871	1,067	3,933
LSTM	4,901	99	4,353	647
BiLSTM	4,657	343	852	4,148

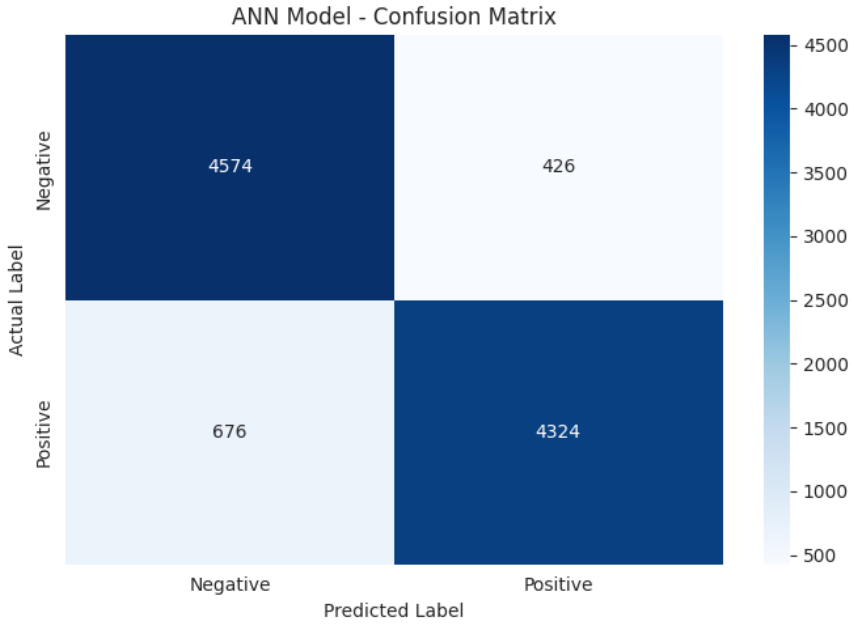


Figure 15: ANN confusion matrix.

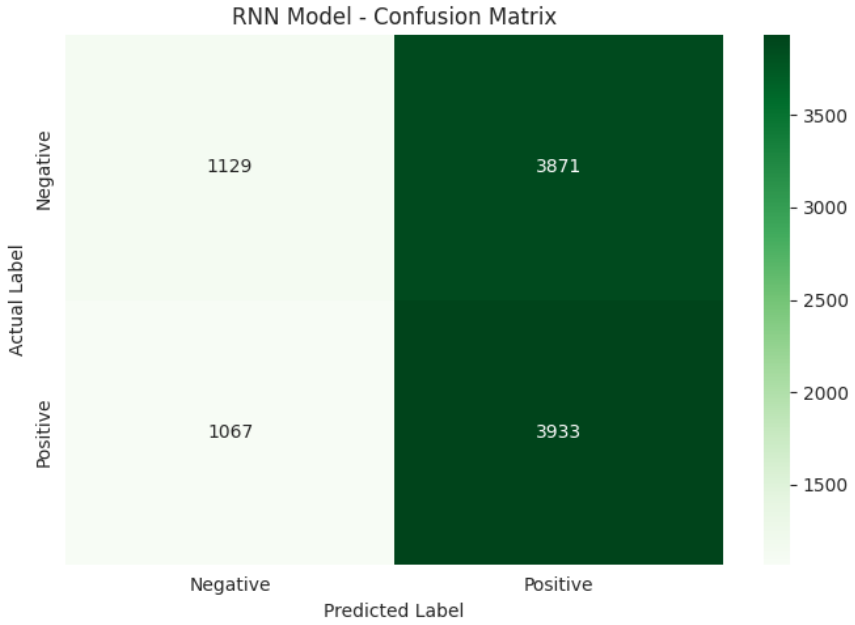


Figure 16: SimpleRNN confusion matrix.

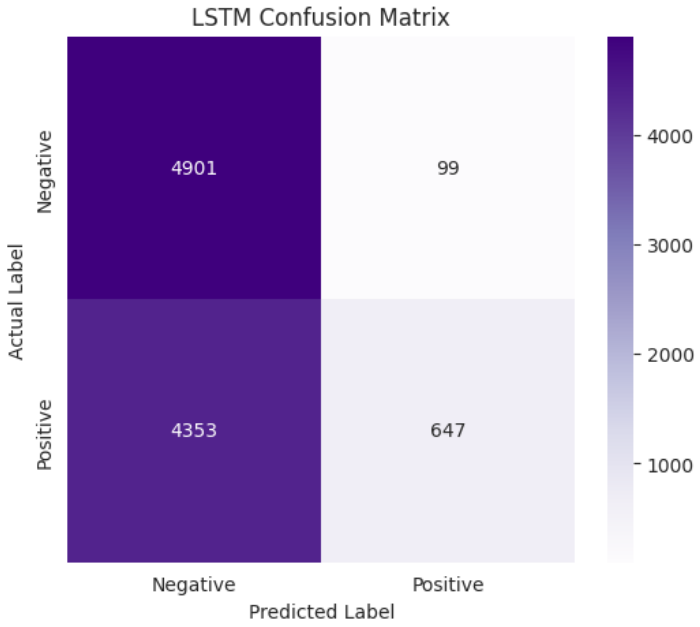


Figure 17: LSTM confusion matrix.

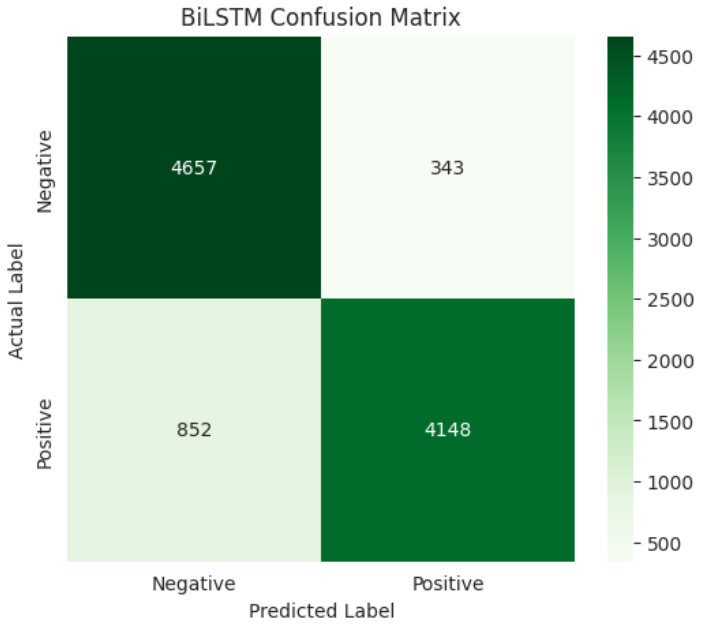


Figure 18: BiLSTM confusion matrix.

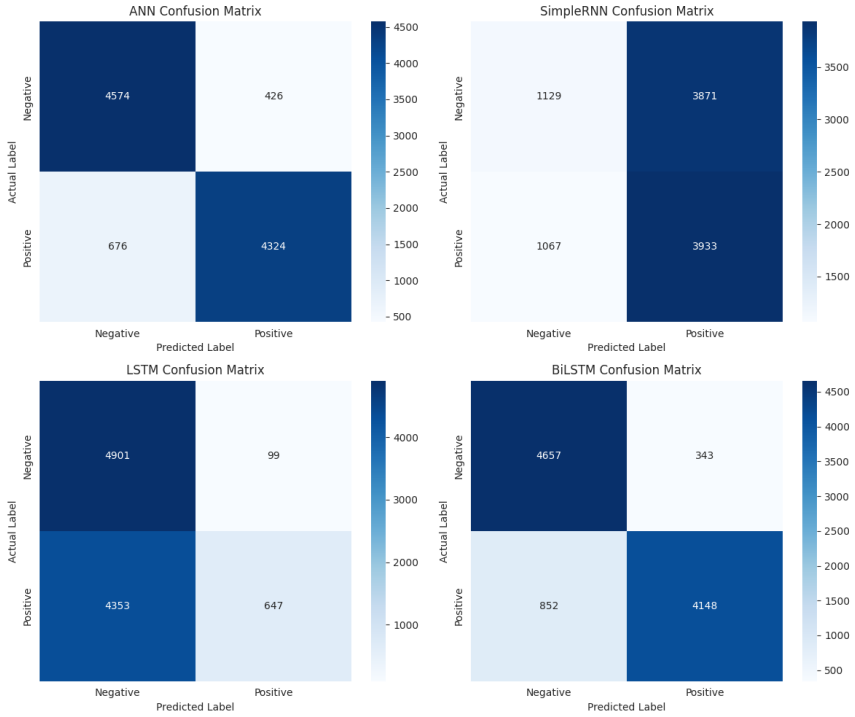


Figure 19: Combined confusion matrix comparison.

12.2.5 Prediction Distribution

The distribution of predicted sentiment categories on the testing set is shown in Table 12.

Table 12: Distribution of predicted sentiment categories.

Model	Negative Predictions	Positive Predictions
ANN	5,250	4,750
SimpleRNN	2,196	7,804
LSTM	9,254	746
BiLSTM	5,509	4,491

12.3 Appendix C: Interpretation of the Experimental Results

Overall performance. The experimental results show that ANN achieved the highest accuracy of 88.98% and the highest F1-score of 88.70%. BiLSTM achieved an accuracy of 88.05% and an F1-score of 87.41%. It also achieved the highest precision of 92.36%. These results indicate that ANN and BiLSTM provided substantially better overall classification performance than SimpleRNN and LSTM under the selected experimental configurations.

SimpleRNN. SimpleRNN achieved an accuracy of 50.62%, indicating weak overall classification performance in this experiment. It produced 3,871 false positives, indicating that it frequently classified negative reviews as positive.

LSTM. LSTM achieved an accuracy of 55.48% and a particularly low recall of 12.94%, showing that it failed to identify many positive reviews. It produced 4,353 false negatives, demonstrating its difficulty in identifying positive reviews.

Confusion matrix analysis. The ANN model correctly classified 4,574 negative reviews and 4,324 positive reviews. It produced 426 false positives and 676 false negatives. BiLSTM correctly classified 4,657 negative reviews and 4,148 positive reviews, with 343 false positives and 852 false negatives. These results indicate that ANN and BiLSTM provided more balanced classification performance than SimpleRNN and LSTM.

Prediction distribution. ANN and BiLSTM produced relatively balanced prediction distributions. SimpleRNN predicted a considerably larger number of positive reviews, while LSTM predicted a considerably larger number of negative reviews. These prediction patterns are consistent with the classification errors observed in their confusion matrices.

Discussion. The experimental results demonstrate that a more complex neural network architecture does not necessarily produce better sentiment classification performance. Although LSTM and BiLSTM are designed to capture sequential information, ANN achieved the best overall performance in this experiment. One possible explanation is that the embedding and Global Average Pooling approach used by ANN was effective for this particular dataset and preprocessing configuration. Global Average Pooling creates a fixed-size representation by averaging word embeddings, although it does not preserve word order.

The poor performance of SimpleRNN and LSTM suggests that their architectures and training configurations may require further optimization. BiLSTM performed strongly, particularly in precision, demonstrating that bidirectional sequence processing can be useful for sentiment classification.

Overall, ANN was the best-performing model based on accuracy and F1-score, while BiLSTM demonstrated the highest precision. These findings are specific to the experimental setup and should not be generalized to all sentiment analysis tasks. The limitations that should be considered when interpreting these results are described in Section 7.

12.4 Appendix D: Supplementary Technical Information

Model parameters. The total parameter counts reported by the four models are shown in Table 13. BiLSTM had the largest number of parameters, while ANN had the smallest number among the four evaluated architectures. Because each model uses a 20,000-word vocabulary and a 64-dimensional embedding, the Embedding layer contributes 1,280,000 parameters (20,000 × 64) to every model; the remaining parameters, derived by subtracting this value from the reported totals, belong to the other layers.

Table 13: Parameter counts of the four models.

Model	Embedding Layer	Other Layers	Total Parameters
ANN	1,280,000	4,225	1,284,225
SimpleRNN	1,280,000	8,321	1,288,321
LSTM	1,280,000	33,089	1,313,089
BiLSTM	1,280,000	66,177	1,346,177

Implementation environment. The research implementation was performed using Python in Google Colab. The notebook `IMDb_50K_Internship_Research.ipynb` contains the complete workflow, including data loading, preprocessing, exploratory data analysis, model development, model training, evaluation, and visualization. The trained models were saved in the Keras format for later reuse.

12.5 Appendix E: List of Abbreviations

Abbreviation	Full Form
ANN	Artificial Neural Network
BiLSTM	Bidirectional Long Short-Term Memory
CNN	Convolutional Neural Network
EDA	Exploratory Data Analysis
FN	False Negative
FP	False Positive
GRU	Gated Recurrent Unit
HTML	HyperText Markup Language
IMDb	Internet Movie Database
LSTM	Long Short-Term Memory
NLP	Natural Language Processing
OOV	Out-of-Vocabulary
ReLU	Rectified Linear Unit
RNN	Recurrent Neural Network
TN	True Negative
TP	True Positive
URL	Uniform Resource Locator